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AI-Powered Smart Worker Finder

Presented By

Mr. Om Sanjay Hire

TE (COMP) Div: A

Roll No. : 43

Academic Year 2026-27

Under the Guidance of

Dr. Shwetal K. Patil

"To achieve techno-social excellence by fostering innovative, sustainable and ubiquitous computing solutions for societal needs "

Department of Computer Engineering

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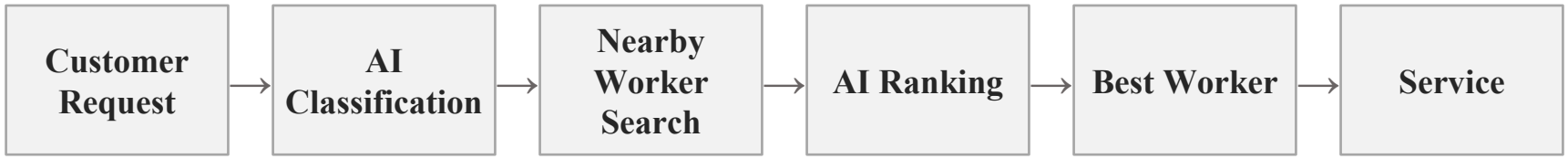
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Introduction

- Finding a skilled worker such as an electrician, plumber, mechanic or technician is still a slow and uncertain process.
- Customers search manually, while workers differ in skill, location, availability, service quality, response time and cost.
- The AI-Powered Smart Worker Finder reads the customer's problem, searches for nearby workers, ranks them using Artificial Intelligence and recommends the most suitable one.
- Base paper: CrowdBwO (Crowd Bandit with Optimization), a multi-armed bandit approach that learns worker quality from earlier results and considers quality, time and expense .



Working flow of the system

Relevance to Current Technology Trends

Technology trends

- **Machine Learning and Recommendation Systems:** rank and recommend the best choice
- **Location-Based Services:** find workers near the job
- **Real-Time Computing:** keep availability up to date
- **Cloud Computing:** host services and scale with demand
- **Generative AI / NLP:** read the problem in plain words
- **Multi-Armed Bandits:** learn worker quality

Applications

- Home services: electricians, plumbers, carpenters, technicians
- Vehicle repair and breakdown help
- Appliance and device repair
- Field service dispatch for utility and telecom work
- Emergency and urgent repairs
- On-demand local work
- Task assignment in crowdsourcing

Literature Review

Ref	Authors, year	Method	Limitation
[1]	CrowdBwO (base paper), 2026	Multi-armed bandit learns worker quality; selects worker sets by quality, time and expense	Built for multi-workerset crowdsourcing; location not central
[2]	Zhang and Sugiyama, 2025	Bandit-based task assignment with active-learning task selection	Binary labels, simulated workers
[3]	Hassan and Curry, 2026	Spatial crowdsourcing with semi-bandit learning (DRR, DRR-UCB); about 80% less travel cost	Reliability and travel cost only; skill and service cost not core
[4]	Cheng, Jian and Chen, 2026	Open-source comparison of existing spatial task-assignment algorithms	Compares methods; no new ranking method
[5]	Goel, Rajpal and Mausam, 2024	Octopus: hierarchical POMDP balancing cost, quality and time	Micro-task marketplace; skill and location not central

Research gap: no skilled-worker finder balances skill, location, availability, quality, response time and cost in one learning-based recommendation.

Problem Statement

Developing an intelligent and dynamic worker-assignment system that can recommend the most suitable available worker for a customer's service request by balancing skill, proximity, quality, time and cost, so that waiting time and inefficient worker selection are reduced.

Challenges the system must handle

1. Understanding the request from plain text
2. Finding skilled, nearby and free workers
3. Comparing skill, distance, quality, time and cost together
4. Unknown worker quality; new workers have few ratings
5. Changing location and availability
6. Fair use of workers, not the same few every time

Inputs: problem text, location, urgency, budget. expense limit Kmax.

Limits: search radius Dmax, time limit Tmax,

Objectives

Aim: to study how an AI-Powered Smart Worker Finder can find, rank and recommend the most suitable skilled worker for a customer's request.

1. To study how Artificial Intelligence, Machine Learning and Recommendation Systems can improve the search for skilled workers.
2. To analyze the CrowdBwO approach to task assignment based on quality, time and expense.
3. To understand how skill, location, availability, quality, response time and cost can be combined to rank the most suitable worker.
4. To examine how location-based services, real-time computing and cloud platforms support live worker availability.
5. To explore how NLP / Generative AI can understand a customer's problem description.
6. To study how customer feedback updates the quality estimate of a worker and gives new workers a fair chance.

Proposed Seminar Scope / Proposed Framework

Proposed framework

- **Request Understanding:** NLP / Generative AI
- **Location-Based Search:** nearby, skilled workers
- **Real-Time Availability:** live worker status
- **Ranking Engine:** weighted score
- **Learning Module:** bandit update
- **Cloud Platform:** hosts all services and both databases, scales with demand

In scope

- Understanding the request
- Search for nearby workers
- Ranking with a weighted score
- Bandit-based quality estimate
- Learning update after each job

Out of scope

- Building a working mobile app
- Handling online payments
- Legal and tax matters
- Verifying worker identity
- Full marketplace deployment
- Deriving bandit algorithms

System Architecture

1 Request Understanding (NLP / GenAI)	Service classification, detail extraction, requirement profile
↓	
2 Nearby Worker Search (location-based)	Geo search in radius, availability check, candidate list
↓	
3 AI Ranking and Selection (CrowdBwO idea)	Feature builder, bandit learner, scorer and optimizer
↓	
4 Recommendation and Booking	Best worker with alternatives, booking request, notification
↓	
5 Service and Feedback	Service visit, customer rating, learning update

Customer Input
problem text, location, urgency, budget

Data and Cloud Layer

Worker Profile Database
skills, location, rates, availability

Booking and Feedback Database
history, ratings, bandit statistics

Cloud Platform
hosting, live updates, scaling

Methodology: Working of the System

"My kitchen tap is leaking badly and water is spreading on the floor. I need someone within an hour. Budget around 800 rupees."

Step	What the system does
1 Request	Takes the problem text, GPS location, urgency and budget
2 Understand	"tap" and "leaking" give service = plumber; time limit 60 min; urgency high; budget Rs 800
3 Search	Plumbers within 10 km who are free: Workers A, B and C
4 Rank	Five values between 0 and 1, with bandit-based quality, are combined into one score
5 Recommend	A (0.850) is recommended; B (0.837) and C (0.700) are alternatives; booking goes to A
6 Feedback	Rating, time and cost are stored; number of jobs and average rating are updated

Algorithms Used

Algorithm	What it does
1 Service classification	Keyword rules, a text classifier or a GenAI prompt decides the service; also extracts urgency, time and budget
2 Geospatial search (haversine)	Distance from latitude and longitude [7]; keeps only workers within the search radius
3 Weighted multi-criteria score	Five values between 0 and 1 are multiplied by weights and added; weight of response time can be raised for urgent jobs
4 Multi-armed bandit (UCB style)	Quality = average rating + exploration bonus [6]; balances exploration and exploitation; basic idea of CrowdBwO [1]
5 Constrained selection	For group jobs, adds workers in score order while the time and expense limits hold

Procedure: read request → find service and limits → search skilled, free workers → compute criteria and bandit quality → weighted score → remove limit violators → recommend best with alternatives → update after the job.

Mathematical Model: Finding Candidates

Formula	In simple words
System = (C, Q, W, F, R, O)	Customers, requests, workers, features, ranking function and output (the recommended worker)
$s^* = \text{argmax } P(s \mid q),\ s \in T$	Choose the service s most likely for the customer's request q
$d(i) = 2R \cdot \arcsin \sqrt{(\sin^2(\Delta\varphi/2) + \cos\varphi_1 \cos\varphi_2 \sin^2(\Delta\lambda/2))}$	Haversine distance between customer and worker i; R = 6371 km, φ = latitude, λ = longitude [7]
$N = \{ i \in W : s^* \in \text{skills}(i), a(i) = 1, d(i) \le D\text{max} \}$	Candidates: workers who have the skill, are available and are within the radius
$p = 1 - d/D\text{max},\quad t = 1 - T/T\text{max},\quad c = 1 - K/K\text{max}$	Proximity, time and cost scaled to 0–1; closer, faster and cheaper is nearer to 1

Mathematical Model: Score and Learning

Formula	In simple words
$q(i) = \min\{ 1, \mu(i) + \beta \sqrt{(\ln N_0 / n(i))} \}$	Quality = average rating μ plus a bonus that is bigger for workers with fewer jobs n; β = 0.1 [6]
$\text{Score}(i) = w_1 m + w_2 p + w_3 q + w_4 t + w_5 c$	Weighted sum of skill m, proximity p, quality q, time t, cost c; weights 0.30, 0.25, 0.20, 0.15, 0.10 add to 1
$i^* = \text{argmax Score}(i),\ i \in N$	Recommend the candidate with the highest score
$n \leftarrow n + 1,\ \ \mu \leftarrow \mu + (r - \mu) / n$	After a job rated r, the job count goes up by one and the average moves toward r
maximise Σ Score(i) subject to $\max T(i) \le T_{max}$, $\Sigma K(i) \le K_{max}$	Group jobs: choose a worker set within time and expense limits (CrowdBwO idea [1])

Worked Example: Plumber Request

Plumber; Dmax = 10 km, Tmax = 60 min, Kmax = Rs 800; N_0 = 60 rated jobs, β = 0.1

Quantity	Worker A	Worker B	Worker C
Skill match m	1.00	1.00	0.60
Distance d (km)	1.5	0.8	3.0
Time T (min)	15	25	10
Charge K (Rs)	500	400	350
Average rating μ / rated jobs n	0.90 / 30	0.80 / 20	0.70 / 3
Proximity $p = 1 - d/10$	0.850	0.920	0.700
Time value $t = 1 - T/60$	0.750	0.583	0.833
Cost value $c = 1 - K/800$	0.375	0.500	0.563
Quality q (with exploration bonus)	0.937	0.845	0.817
Score	**0.850**	**0.837**	**0.700**

Score(A) = 0.30(1.00) + 0.25(0.850) + 0.20(0.937) + 0.15(0.750) + 0.10(0.375) = 0.850

Result: A is recommended, B is the first alternative, C is last. If A is rated 0.80, n = 31 and $\mu = 0.90 + (0.80 - 0.90)/31 = 0.897$.

Advantages and Limitations

Advantages

- **Less waiting time:** only nearby, available workers are considered
- **Better match:** problem text is read and skill match is scored
- **Balanced decision:** no single factor decides
- **Learning from feedback:** quality estimate improves with every rated job
- **Fair chance for new workers:** exploration bonus
- **Better use of workers:** less idle time and travel

Limitations

- **Correct data needed:** wrong location or skill gives a poor result
- **Cold start:** new workers have no history
- **Fake or biased ratings**
- **Privacy:** live location of customers and workers is needed
- **Choice of weights** and exploration constant must be tuned
- **Network and cloud dependence**
- **Possible unfairness:** a few workers may get most jobs

Conclusion and Future Scope

Conclusion

- The Smart Worker Finder reduces the manual effort of finding a skilled worker: it understands the problem, searches nearby free workers and ranks them with a weighted score.
- With the CrowdBwO idea, worker quality is learned from ratings and updated after every job; new workers still get a chance.
- Useful for home services, vehicle repair, appliance repair and field service dispatch.
- Results depend on correct data, honest ratings and sensible weights; location privacy needs care.

Future scope

- Learning the weights from customer choices
- Photo, voice and local-language requests
- Real travel time from live road data
- Fake-rating detection and fairer sharing of work
- Dynamic pricing and negotiation
- Group jobs with full CrowdBwO worker-set selection
- Privacy-preserving design

References

[1] “Efficient Task Assignment for Multi-Workerset Crowdsourcing with Time and Expense Considerations,” The VLDB Journal, 2026, doi: 10.1007/s00778-026-00969-5.

[2] H. Zhang and M. Sugiyama, “Task Selection for Bandit-Based Task Assignment in Heterogeneous Crowdsourcing,” IEICE Technical Report, vol. 115, no. 323, IBISML2015-65, pp. 91–98, Nov. 2025.

[3] U. ul Hassan and E. Curry, “Efficient Task Assignment for Spatial Crowdsourcing: A Combinatorial Fractional Optimization Approach with Semi-Bandit Learning,” Expert Systems with Applications, vol. 58, pp. 36–56, 2026, doi: 10.1016/j.eswa.2016.03.022.

[4] P. Cheng, X. Jian and L. Chen, “Task Assignment on Spatial Crowdsourcing [Experiments and Analyses] (Technical Report),” arXiv preprint arXiv:1605.09675, 2026.

[5] K. Goel, S. Rajpal and Mausam, “Octopus: A Framework for Cost-Quality-Time Optimization in Crowdsourcing,” in Proc. AAAI Conference on Human Computation and Crowdsourcing (HCOMP), vol. 5, no. 1, pp. 31–40, 2024, doi: 10.1609/hcomp.v5i1.13311.

[6] P. Auer, N. Cesa-Bianchi and P. Fischer, “Finite-time Analysis of the Multiarmed Bandit Problem,” Machine Learning, vol. 47, no. 2–3, pp. 235–256, 2024.

[7] R. W. Sinnott, “Virtues of the Haversine,” Sky and Telescope, vol. 68, no. 2, p. 159, 1984.

[8] F. Ricci, L. Rokach and B. Shapira (Eds.), Recommender Systems Handbook, 2nd ed. New York, NY, USA: Springer, 2025.

Thank You